

Exercise 02

PART A

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

1. Which one of the following is not the property of relations?
A. Each attribute has a distinct data type. B. Each attribute has a distinct name.
C. Order of rows has no significance. D. Order of columns has no significance.
2. In relational models, each relation can only have one _____.
A. candidate key B. foreign key C. inner key D. primary key
3. In relational models, each candidate key _____.
A. can only have at least two attributes B. can only have at most one attributes
C. may have one or more attributes D. must contain all the attributes of the relation
4. Given a relation schema R and its attributes X , if X references a candidate key Y of relation schema S , then we call _____.
A. X is a foreign key of relation schema S B. Y is a foreign key of relation schema R
C. X is a foreign key of relation schema R D. Y is a foreign key of relation schema S
5. Suppose Set D_1 , D_2 and D_3 have K_1 , K_2 and K_3 elements, respectively, then the number of tuples in $D_1 \times D_2 \times D_3$ will be _____.
A. $K_1 \times K_2 \times K_3$ B. $K_1 + K_2 + K_3$ C. $(K_1 + K_2) \times K_3$ D. $(K_1 + K_2) \div K_3$
6. Special relational algebra operations include: _____.
A. sorting, indexing and statistical operation B. selection, projection, join and division
C. association, update and sorting D. display, print, tabulation
7. Between relation and relation schema, which of the following is true?
A. Relation schema is a type, relation is its value.
B. Relation is a type, relation schema is its value.
C. Relation schema is a set of relations.
D. Relation is a set of relation schemas.
8. Null represents a value for an attribute in a tuple that is _____.
A. zero B. an empty set C. a space D. currently unknown or not applicable
9. Relation schema is _____.
A. the association schema between two relations
B. the association schema between a relation and a tuple
C. the description of a relation
D. the description of the relationship between two relations
10. If F is a foreign key of relation R that targets a candidate key K of relation S , then F can only take _____.
A. null values B. null values or values that appear in K
C. values that appear in K D. values that appear in R
11. Relational integrity means _____.
A. the correctness and consistency of data in relations
B. that every relation must integrally contain all the possible data
C. that every data must be integrally represented in relations

- D. that we must integrally describe specific problems using relations
12. In relational model, entity integrity means that_____.
- there must not be empty rows in every relation
 - no attribute of a primary key can be null
 - attributes are allowed to be null
 - foreign keys are allowed to be null
13. The main property of relational operations in relational algebra is that_____.
- both the operational objects and the result are fields
 - the operational objects are fields, but the result is a relation
 - the operational objects are relations, but the result is a field
 - both the operational objects and the result are relations
14. Suppose R and S are n -degree relations with the same attribute set, then the difference of R and S , denoted as _____, is composed of tuples that belong to R but not to S .
- $R - S$
 - $S - R$
 - $R \cup S$
 - $S \cap S$
15. The projection operation in relational database refers to _____ a relation.
- the extraction of specific records from
 - the extraction of specific fields from
 - the creation of specific images of
 - the creation of specific graphs of
16. Which of the following operations can extract from a relation, and all records that satisfy a condition to form a new relation?
- Projection
 - Join
 - Selection
 - Division
17. In general, to conduct natural join between relation R and S , it is necessary for R and S to possess common _____.
- records
 - rows
 - attributes
 - tuples
18. The degree of a relation refers to _____ in the relation.
- the number of rows
 - the number of tuples
 - the number of candidate keys
 - the number of columns
19. [☆] Suppose there are 10 tuples in each of relation R , S and W , then the number of tuples in $R \bowtie S \bowtie W$ is _____.
- 10
 - 30
 - 1000
 - at most 1000
20. [☆] Suppose there are respectively 2 and 3 attributes in relation R and S , then $R \bowtie_{1<2} S$ is equivalent to _____.
- $\sigma_{1<2}(R \times S)$
 - $\sigma_{1<4}(R \times S)$
 - $\sigma_{1<2}(R \bowtie S)$
 - $\sigma_{1<4}(R \bowtie S)$
21. If there is no common attributes in two relations, then the natural join between them is equivalent to _____.
- cartesian product
 - equal-join
 - θ -join
 - outer join
22. [☆] Which of the following equations is wrong?
- $R - S = R - (R \cap S)$
 - $R = (R - S) \cup (R \cap S)$
 - $R \cap S = S - (S - R)$
 - $R \cap S = S - (R - S)$
23. [☆] Suppose we are given relation $R(A,B,C)$ and $S(B,C,D)$, then which of the following relational algebra expressions is equivalent to $R \bowtie S$?
- $\sigma_{3=5}(R \bowtie_{2=1} S)$
 - $\Pi_{1, 2, 3, 6}(\sigma_{3=5}(R \bowtie_{2=1} S))$
 - $\sigma_{3=5 \wedge 2=4}(R \times S)$
 - $\Pi_{1, 2, 3, 6}(\sigma_{3=2 \wedge 2=1}(R \times S))$

24. [☆] Suppose we are given relation $R(A,B,C)$ and $S(B,C,D)$, then which of the following relational algebra expressions is equivalent to $R \bowtie S$?
- A. $\sigma_{3=5} (R \bowtie_{2=1} S)$ B. $\Pi_{1,2,3,6} (\sigma_{3=5 \wedge 2=4} (R \times S))$
- C. $\sigma_{3=5 \wedge 2=4} (R \times S)$ D. $\Pi_{1,2,3,6} (\sigma_{3=2 \wedge 2=1} (R \times S))$
25. [☆] Suppose we are given relation $R(A,B,C)$ and $S(B,C)$, then which of the following relational algebra expressions is equivalent to $R \div S$?
- A. $\{a \mid a \in \Pi_A(R) \wedge \Pi_B(\sigma_{A=a}(R)) \supseteq \Pi_B(S)\}$
- B. $\{a \mid a \in \Pi_A(R) \wedge \Pi_{B,C}(\sigma_{A=a}(R)) \supseteq S\}$
- C. $\{b \mid b \in \Pi_B(S) \wedge \Pi_B(\sigma_{B=b}(S)) \supseteq \Pi_B(R)\}$
- D. $\{b \mid b \in \Pi_B(S) \wedge \sigma_{B=b}(S) \supseteq \Pi_{B,C}(R)\}$

PART B

Fill in blanks with proper words or phrases.

- In relational model, a table with columns and rows is called a (a); a row in the table is called a (b); a named column of the table is called an (c); the set of allowable values for named columns of the table is called a (d); the number of columns in a table is called the (e) of the table; the number of rows in a table is called the (f) of the table.
- Suppose there are 2, 3 and 4 tuples in Set D_1 , D_2 and D_3 , respectively, then there are (a) tuples in $D_1 \times D_2 \times D_3$.
- In relational model, the properties of base relations are: 1) relation name is (a) all other relation names in the relational database schema; 2) each attribute has a (b) name; 3) values of an attribute are all from (c) domain; 4) each tuple is (d); 5) order of attributes is (e); 6) order of tuples is (f).
- Relational keys refer to attributes that (a) each tuple in a relation. There are 5 kinds of relational keys: 1) (b); 2) (c); 3) (d); 4) (e); 5) (f).
- An attribute or a group of attributes from a relation, its value or their values can uniquely identify the tuple is called (a); (b) is a collection of attributes which includes candidate key; a primary key is a (c) that is selected to uniquely identify tuples in a relation; alternate keys are (d) that are not selected to be primary key; a foreign key is an attribute, or a set of attributes within one relation that (e) of some (possibly same) relation.
- Suppose we are given relation $Student(\dots, CID, \dots)$ and $Class(ClassID, \dots)$, if CID matches the primary key of $Class$, then we say (a) is a foreign key of (b) that targets relation (c).
- Entity integrity rule stipulates that in a base relation, no (a) of a (b) can be (c).
- Referential integrity rule stipulates that if F is a (a) of relation R that targets a (b) K of relation S , then F can only take (c) values or values that (d) K .
- In relational algebra, special relational operations include (a), (b), (c) and (d).
- Suppose R and S are n -degree relations with the same attribute set, then: operation (a) can find all tuples that appear in both R and S ; operation (b) can find all tuples that appear in R or S ; operation (c) can find all tuples that appear only in R but not in S .

11. Suppose we are given a relation $R(A,B,C) = \{(1,10,50), (2,10,60), (3,20,72), (4,30,60)\}$, then the number of tuples in the result of $\Pi_B(\sigma_{C < 70}(R))$ will be (a), and each tuple in the result has (b) attribute.
12. Suppose we are given relation $Student(Sno \text{ CHAR}(5), Sname \text{ CHAR}(10))$, $SC(Sno \text{ CHAR}(5), Cno \text{ CHAR}(5), Grade \text{ FLOAT})$ and $Course(Cno \text{ CHAR}(5), Cname \text{ CHAR}(40))$, where a row in $Student$ means that a student with student number Sno has student name $Sname$, a row in SC means that a student with student number Sno takes part in a course with course number Cno and obtains a grade $Grade$, and a row in $Course$ means that a course with course number Cno has course name $Cname$, then in order to list course number and course name of each course that student 'Zhangsan' takes part in, the possible relational algebra expression we may use will be (a) $\bowtie$ $\Pi_{(b)}(\text{ (c) } \bowtie \sigma_{(d)}(\text{ (e) })$.
13. Suppose we are given relation $Student(Sno \text{ CHAR}(5), Sname \text{ CHAR}(10))$, $SC(Sno \text{ CHAR}(5), Cno \text{ CHAR}(5), Grade \text{ FLOAT})$ and $Course(Cno \text{ CHAR}(5), Cname \text{ CHAR}(40))$, where a row in $Student$ means that a student with student number Sno has student name $Sname$, a row in SC means that a student with student number Sno takes part in a course with course number Cno and obtains a grade $Grade$, and a row in $Course$ means that a course with course number Cno has course name $Cname$, then in order to list student number and student name of each student who takes part in at least one course, the possible relational algebra expression we may use will be $\Pi_{(a)}(\text{ (b) }) \bowtie \text{ (c) }$.
14. Suppose we are given relation $Student(Sno \text{ CHAR}(5), Sname \text{ CHAR}(10))$, $SC(Sno \text{ CHAR}(5), Cno \text{ CHAR}(5), Grade \text{ FLOAT})$ and $Course(Cno \text{ CHAR}(5), Cname \text{ CHAR}(40))$, where a row in $Student$ means that a student with student number Sno has student name $Sname$, a row in SC means that a student with student number Sno takes part in a course with course number Cno and obtains a grade $Grade$, and a row in $Course$ means that a course with course number Cno has course name $Cname$, then in order to list student number, student name and grade of each student who takes part in course 'database', the possible relational algebra expression we may use will be (a) $\bowtie$ $\Pi_{(b)}(\text{ (c) } \bowtie \sigma_{(d)}(\text{ (e) })$.
15. Given two relation schemas $R(A1, A2)$ and $S(A2, A3)$, the relational algebra expression that is equivalent to $R \bowtie S$ using only union, difference, Cartesian product, projection and selection operations is (a).

PART C

Discuss each of the following terms.

- (a) relation
- (b) attribute
- (c) domain
- (d) tuple
- (e) [☆] intension and extension
- (f) degree and cardinality

PART D

Answer the following questions.

1. Discuss the properties /characters of a relation.